

ANSH MANI TRIPATHI

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EDUCATION

SRM Institute of Science and Technology, Kattankulathur	2023 – 2027
B.Tech in Computer Science and Engineering – Gaming Technology	CGPA: 9.4 / 10
St. Fidelis College, Lucknow	2020 – 2022
Class XII (ISC) – 89.25% Class X (ICSE) – 92.67%	

TECHNICAL SKILLS

Backend	Node.js, Bun, Python, REST APIs, WebSockets (Socket.io), PostgreSQL, Supabase, Docker, Kubernetes
Frontend	React.js, TypeScript, Vite, Tailwind CSS, Swift (SwiftUI), HTML5 Canvas
Other	C#, Java, JavaScript, Unity, Unreal Engine, BeautifulSoup, Git, Vercel, Nginx
Languages	English (Professional), Hindi (Native)

WORK EXPERIENCE

iOS App Developer Intern – Infosys	March 2026
Built Ochima, a full-stack fleet management app in a team of 10; engineered the server in Django (Python) with RESTful APIs handling fleet data, vehicle state, and operator workflows, and the iOS frontend in Swift.	

PROJECTS

Box Game Distributed Multiplayer Web Game	Feb 2026 – Present
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Personal Project | Node.js, TypeScript, Bun, Socket.io, HTML5 Canvas, Oracle Cloud, Kubernetes, Docker, Nginx

- Re-architected the game from a single monolithic server into a horizontally scalable distributed system – a **Gateway** for matchmaking, an orchestrating **Server Manager**, and a fleet of self-registering game-server pods – routing each new room to the least-loaded server via a custom **min-heap** scheduler ($O(\log n)$ assign).
- Built the authoritative real-time game server in **Node.js/Socket.io** with delta-state networking (only changed state broadcast per tick) and spatial-grid collision detection for $O(1)$ platform lookups, supporting rooms of up to 20 concurrent players at a configurable tick rate; wrote the Gateway and Server Manager in **TypeScript on Bun**.
- Self-hosted the entire fleet on an Oracle Cloud VM**, running **Kubernetes** with hostNetwork pods and RBAC-scoped dynamic pod creation, and terminated TLS with a host-level **Nginx stream** proxy that maps 1,000 external wss ports to internal game-server ports for browser-compatible WebSocket connections.

The Industrial Chemist iOS Learning App	2025 – 2026
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Academic Project | Swift, UIKit, Firebase Auth, Cloud Firestore, StoreKit, React, TypeScript, Vite

- Built a full-featured iOS app (team of 3) that turns industrial chemistry processes into gamified virtual experiments; users progress through a four-phase flow (Theory → Build → Test → Results) across five experiments including the Haber, Ostwald, and Contact processes, with embedded **USDZ 3D models** for molecules and apparatus.
- Implemented an **XP and division system** (Bronze through Master) with daily streaks, a live **Firestore-backed leaderboard**, community posts, and a **StoreKit** subscription tier unlocking premium experiments; shipped a companion marketing website in **React + TypeScript + Vite**.

WorkNovasLLC – Taki Candidate Submission Platform	March 2026 – Present
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Freelance | React + TypeScript, Node.js, Express.js, PostgreSQL, WebSockets, JWT

- Designed the **PostgreSQL** schema with role-scoped access control across three tiers (Admin, Account Manager, Recruiter); admins create JDs, AMs distribute them, recruiters submit candidates.
- Implemented **real-time WebSocket notifications** so admins receive instant alerts on new submissions without polling, and secured all routes with **JWT-based auth** with role-gated middleware.

WorkNovasLLC – Billing Payment Management System	March 2026 – Present
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Freelance | React + Vite, Node.js, Express.js, PostgreSQL, Vercel, Supabase

- Built RESTful backend APIs with **Node.js/Express.js** for invoice creation, payment tracking, and billing history, backed by **PostgreSQL**; deployed frontend on **Vercel** and database on **Supabase**.

Srikanth Nursery & Lakshmi Associates Business Website	Oct 2025 – Apr 2026
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Freelance | React + Vite, Tailwind CSS, Node.js, Express.js, Supabase (PostgreSQL)

- Built full-stack site with **Node.js/Express.js** backend and **Supabase PostgreSQL**; exposed RESTful APIs for portfolio, services, and client data with server-side category filtering.

ACHIEVEMENTS & INFRASTRUCTURE

OssomeHacks 2.0 – 1st Place (Crypto Domain) Team OutOfFunds	2024
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Apple Swift Student Challenge 2026 – Submitted *Meter Logger*, a SwiftUI app that tracks and splits electricity bills across flats proportionally, with persistent bill history via UserDefaults.

Self-Hosted Minecraft Server | Oracle Cloud VM, Nginx – Deployed a Minecraft server on an **Oracle Cloud VM** with **Nginx** reverse proxying client TCP connections to the game server socket. After the server was compromised through an open port, implemented **IP-based firewall filtering** to restrict access to trusted addresses only.